KIRTAN JAIN

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Education

- **Indian Institute of Technology, Patna** | *B.Tech in Computer Science and Engineering* 2021 – 2025

Work Experience

- **Samsung Research Institute - Noida** Jul 2025 – Present
  R&D Engineer *Noida, India*
  - Built a **Go peer-to-peer** tool (CLI and GUI) that shares multi-GB SDK builds across engineers' machines with **gossip** discovery, **Merkle-tree** sync, and **parallel rsync** streams, making **4-hour** pulls from Korea **3x** faster.
  - Wrote its **Go** backend: a **stateless HTTP** service ingesting JSON telemetry, a **job scheduler** with file-lock crash detection, a **SQL** store with versioned migrations, and a **systemd** daemon, covered by **460+ tests**.
  - Wrote **Python MCP** servers that let AI coding assistants like **Claude Code** use internal build and defect-tracking portals with no API (**QuickBuild**, **PLM**), cutting manual steps from daily triage.
  - Designed and built **AutoIVR**, an **agentic**, cross-language IVR feature: **speech-to-text** reads the helpline menu on the handset, an **on-device LLM** picks the caller's option, and the dialer presses keys until a person answers.
  - Land **Java** and **Kotlin** changes in **Android Framework WindowManager** (split screen, freeform, **DeX**) for **One UI 8.5 and 9.0**, root-causing defects from dumpstate and logcat, and raising **AOSP** issues when the cause is upstream.
  - Invention on adaptive accessibility **approved by Samsung IP review**, pending filing as a patent.
- **Sprinklr** May 2024 – Jul 2024
  Product Engineering Intern *Gurgaon, India*
  - Built middleware with a **custom control UI** that captures **GraphQL/REST** responses and replays them as mocks, so frontend work continued through backend outages, released as an **npm package** other teams adopted.

Projects

- **Alloc8 - Room Allocation Portal** | *Node.js, MongoDB, Redis* Website
  Project Lead *Apr 2024 – Jun 2024*
  - Led a portal on **AWS** that placed **3,000 students** into 1,400 hostel rooms in **minutes**, using **Redis distributed locks** so no room was ever double-booked and roommate pairs committed together or not at all.
- **Land Records** | *Node.js, Python, Playwright, SQLite, Kotlin, Jetpack Compose* GitHub
  Author *Jun 2026 – Aug 2026*
  - Automated retrieval of **Gujarat** land records from two **ASP.NET WebForms** government portals with no API or bulk export, driving their session, view-state, and **CAPTCHA** flows over **HTTP** from **Node** and **Python**.
  - Solved each portal's **CAPTCHA** automatically, pulled PDFs via **Playwright**, and caught error pages posing as PDFs.
  - Built a native **Kotlin** and **Jetpack Compose** app storing records in **Room** for offline use, with its own update channel.
- **Spine** | *TypeScript, Cloudflare Workers, D1* GitHub
  Author *Jul 2026*
  - **Serverless** KOReader sync server on **Cloudflare Workers** and **D1** implementing the stock **kosync** protocol, so clients need only a URL change, with per-device progress and a nightly **cron** rollup.

Publications

- **When Words Can't Capture It All: Video-Based User Complaint Generation** arXiv
  CIKM 2025 *Dec 2024 – May 2025*
  - Defined **CoD-V**, turning a video of a fault into a written support report, curated **ComVID** (1,176 videos), and fine-tuned **VideoLLaMA 2** with multimodal retrieval to beat published captioning baselines.
- **M3Hop-CoT: Misogynous Meme Identification** arXiv
  EMNLP 2024 *Jan 2024 – May 2024*
  - **Chain-of-Thought** method detecting implicit misogyny in memes by reasoning over emotion, target and context.

Technical Skills

Languages: Go, Python, C++, Java, Kotlin, JavaScript, TypeScript, SQL, Bash
Backend & Distributed Systems: HTTP/REST, GraphQL, concurrency, P2P sync, job queues, distributed locks
Databases: MySQL, SQLite, Redis, MongoDB, Cloudflare D1, schema migrations
Cloud, Frameworks & Tools: AWS, Cloudflare Workers, Node.js, Flask, Caddy, Git, Perforce, MCP, LLM APIs

Achievements/Extracurricular

- **ICPC India Regional Finalist 2024** (Amritapuri), top 3 at IIT Patna, top 300 nationally | **Codeforces** Expert (1805) | **CodeChef** 4⋆ (1927).
- **Global rank 427**, **Google Kick Start** | **Round 2**, **Meta Hacker Cup 2023**, 1,357th worldwide | **KVPY Fellow** (SA-2019, SX-2020) | **AIR 2194**, JEE Advanced 2021.
- **General Secretary, Technical**, Student Gymkhana '24-25 | **Overall Events Head**, Inter-IIT Cultural Meet 7.0.